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HypEReg-TransMorph: A Hyperelastic-Regularized Transformer Framework for Folding-Suppressed Deformable Brain MRI Registration

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Abstract

Accurate deformable brain MRI registration demands both high anatomical alignment and geometrically plausible deformation fields, yet contemporary Transformer-based methods routinely produce fields with folded voxels that violate the requirements of downstream analysis pipelines. We integrate hyperelastic-energy regularization (HypEReg)—a differentiable, plug-in training-loss term combining a volume-distortion penalty with an explicit anti-folding hinge—into a TransMorph-style dense registration backbone. On a held-out IXI atlas-to-subject protocol (115 test subjects), evaluated with grouped anatomical Dice, Jacobian-determinant statistics, and paired Wilcoxon/BH-FDR inferential testing, HypEReg-TransMorph achieves a mean grouped-structure Dice of 0.7537—comparable to the unregularized TransMorph baseline (0.7527)—while reducing the fraction of non-positive Jacobian voxels from 1.50% to 0.0015% and improving HD95 from 5.88 mm to 3.16 mm. The improvement in deformation regularity is statistically significant ($p < 10^{-19}$, FDR-adjusted $q < 10^{-19}$, rank-biserial $r = 1.00$). Because HypEReg acts solely at training time, pure forward-pass inference cost remains identical to the baseline (0.0822 s per volume, 5.685 GB peak memory). These results demonstrate that explicit Jacobian-aware regularization can deliver near-diffeomorphic Transformer-based registration without compromising alignment accuracy or inference efficiency, with potential benefit for downstream applications including Jacobian-based morphometric analysis, atlas-based segmentation, treatment-planning support, and longitudinal disease monitoring.

Keywords: medical image registration; deformable registration; transformer; hyperelastic-energy regularization; HypEReg-TransMorph; folding suppression; Jacobian regularization; Jacobian determinant; brain MRI

1. Introduction

Deformable registration is a central component of computational neuroimaging pipelines, including atlas propagation, longitudinal analysis, and voxel-wise morphometry. In recent years, learning-based methods have significantly reduced inference time compared with classical iterative optimization [1,2]. However, high overlap metrics alone are not sufficient when deformation fields contain local foldings or implausible volume changes.

Transformer-based registration improves long-range contextual modeling and has shown strong performance on 3D MRI tasks [3]. Nevertheless, unconstrained displacement

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prediction may still produce non-diffeomorphic deformations, which can invalidate downstream Jacobian-based analysis. To address this, we adapt a Jacobian-aware regularization scheme derived from hyperelastic principles [4] to a TransMorph backbone and explicitly optimize deformation plausibility. We refer to this regularizer as HypEReg, with the explicit caveat that we use a deep-learning–friendly two-term subset (volume distortion and anti-folding) rather than the full polyconvex hyperelastic energy.

Clinical motivation. The need for diffeomorphic, fold-free deformations is not a purely mathematical concern: in clinical neuroimaging, the Jacobian determinant of the registration field is itself the quantity reported in tensor-based morphometry (TBM) and longitudinal atrophy/expansion studies for Alzheimer's disease, mild cognitive impairment, multiple sclerosis, normal-pressure hydrocephalus, and post-surgical follow-up of brain tumors and epilepsy resections. When local foldings ($\det J_\phi \leq 0$) appear in the deformation field, the corresponding voxels report negative or undefined volume change, biasing voxel-wise statistical maps and inflating false-positive clusters at the boundaries of ventricles, cortex and deep gray nuclei—precisely the structures of greatest clinical interest. Equivalently, in atlas-based segmentation (FreeSurfer-style and SPM-style pipelines used for radiotherapy planning, deep brain stimulation targeting, and surgical navigation), label propagation through a folded warp can produce anatomically inconsistent boundaries that must be manually corrected. Constraining the registration field to be near-diffeomorphic at training time therefore directly reduces downstream clinical post-processing burden and improves the trustworthiness of derived biomarkers, while preserving the inference-time speed of feed-forward learned models that is essential for clinical throughput.

The contributions of this study are positioned, deliberately, as engineering-level rather than theoretical:

- An empirical instantiation of a Jacobian-aware regularizer derived from hyperelastic principles inside a Transformer-based dense-displacement registration framework (HypEReg-TransMorph), explicitly distinct from prior CNN/U-Net implementations of related volume-change and anti-folding penalties [5,6] and from concurrent linear-elastic and hyperelastic regularizers in deep registration [7,8];
- A unified IXI atlas-to-subject benchmark that quantifies the trade-off between grouped-VOI overlap and deformation regularity across deep-learning baselines and a classical SyN reference, including paired Wilcoxon/BH-FDR inferential analysis;
- A reproducibility package (code, configurations, per-case metric exports, and figure scripts) released openly so that the regularization design and evaluation protocol can be reused as a drop-in component on TransMorph-style backbones.

2. Related Work

2.1. Classical and Diffeomorphic Registration

Classical registration frameworks such as SyN and related diffeomorphic formulations remain strong references for deformation plausibility, because they explicitly constrain transformation smoothness and invertibility during optimization [9–11]. These methods often provide robust topology behavior but at substantially higher computational cost than feed-forward learned models.

2.2. Learning-Based Dense Registration

Learning-based methods, represented by VoxelMorph and descendants, provide fast inference with strong overlap accuracy [1]. Subsequent CNN and hybrid variants (e.g., CycleMorph and MIDIR) improve representational capacity [2], while probabilistic diffeomorphic variants aim to improve deformation stability [12]. Even so, unconstrained dense displacement prediction can still produce locally implausible Jacobian behavior.

2.3. Transformer-Based Registration

Transformer backbones improve long-range context modeling in volumetric registration and have become competitive baselines [3,13]. In this manuscript, CoTr, nnFormer, and PVT are treated as adapted Transformer-family backbones used in the same IXI evaluation workflow [14–16]. For each backbone, the registration adapter concatenates moving/fixed images as a two-channel input and uses the repository flow-prediction decoder head to output (I_w, ϕ); all three adapters are evaluated under the same atlas-to-subject protocol. The objective here is not to propose a new Transformer architecture, but to evaluate training-time regularization on a TransMorph-style dense-flow network.

2.4. Hyperelastic and Jacobian-Based Regularization

Jacobian-aware and hyperelastic penalties provide direct control of local volume change and folding behavior [4]. These terms complement smoothness regularizers by explicitly penalizing singular or near-singular deformation regions, which is important for downstream morphometric interpretation. The original Burger–Modersitzki–Ruthotto formulation combines length, surface (cofactor) and volume terms inside a polyconvex energy and proves existence of minimizers in a discretize-then-optimize variational setting; in our deep-learning setting, we use only a Jacobian-aware subset of this family, and we discuss this divergence explicitly in Section 3.4.

2.5. Volume-Preserving and Anti-Folding Regularization in Deep Registration

Several recent learning-based works place Jacobian-aware penalties directly into the training objective of deep registration networks. Mok and Chung [5] introduce a selective Jacobian-determinant penalty that activates on $\det J \leq 0$ regions inside a fast symmetric diffeomorphic CNN; this term is conceptually the immediate antecedent of our anti-folding hinge $\mathcal{L}_{\text{fold}}$. Hering et al. [6] place a volume-change penalty together with a curvature regularizer inside a CNN-based lung-CT registration pipeline, demonstrating that BMR-style volume control can be incorporated into deep registration loss functions. Zou et al. [7] adopt a non-linear conformal-invariant hyperelastic energy in a deep registration framework on lung CT, providing the closest precedent for hyperelastic-style regularization in learned registration. The recent *ElasticMorph* framework [8] plugs a second-order Navier–Cauchy linear-elastic regularizer into TransMorph and VoxelMorph backbones and reports overlap improvements with reduced folding artifacts on IXI/LPBA40 brain registration. Compared with these works, the contribution of HypEReg-TransMorph is restricted in scope: we do not propose a new architecture or a new theoretical regularizer; instead we report a practical configuration that combines a simplified rational volume-change term with an explicit anti-folding hinge inside a TransMorph backbone, and we evaluate the resulting trade-off under the IXI atlas-to-subject protocol with paired inferential statistics. We discuss the precise differences in formulation in Section 3.4 and the experimental positioning relative to ElasticMorph in Section 5.

3. Materials and Methods

3.1. Dataset and Preprocessing

Experiments use the IXI brain MRI dataset [17] under an atlas-to-subject registration protocol with a 403/58/115 train/validation/test split. All reported metrics are computed on the 115 held-out test subjects. This study uses the T1-weighted preprocessed IXI package commonly adopted by TransMorph-style studies [2,3], where preprocessing includes skull stripping, affine alignment, and subcortical segmentation generation in the published preprocessing workflow. Volumes are represented in a common template space and cropped/resampled to $160 \times 192 \times 224$. A shared fixed atlas image-label pair is used

for all methods. Intensity images are warped with trilinear interpolation and label maps with nearest-neighbor interpolation in evaluation/training utilities.

Table 1. Dataset, preprocessing, and protocol summary.

Item	Configuration
Dataset source	IXI project dataset [17] and preprocessed IXI package used in TransMorph-style workflows
License	IXI public dataset terms (CC BY-SA 3.0 at IXI portal); preprocessed release follows its repository terms
Volumes and split	576 preprocessed subjects + atlas; 403 / 58 / 115 (train / validation / test)
Modality and task	T1-weighted structural MRI; atlas-to-subject deformable registration
Input shape	$160 \times 192 \times 224$ after preprocessing/cropping
Preprocessing summary	Public preprocessed IXI workflow (including skull stripping, affine alignment, and segmentation preprocessing) with template-space normalization; same pipeline used across all compared methods
Labels and grouped VOIs	Segmentation labels are aggregated into 17 bilateral/related VOI groups for grouped Dice analysis
Atlas definition	Fixed atlas image/label pair; same atlas used for all methods
Interpolation	Intensity: trilinear warping; labels: nearest-neighbor warping
Metric spacing	Surface metrics (HD95/ASSD/NSD) use spacing-aware distance transforms in the evaluation metrics implementation
Compared families	TransMorph-family baselines, CNN/hybrid baselines, and classical SyN

3.2. Baseline Model

The base architecture is TransMorph, a Transformer-based dense deformable registration network with a Swin-style encoder and decoder-to-flow prediction head [3,13]. We compare HypEReg-TransMorph against deep-learning and classical baselines under identical preprocessing and evaluation scripts:

- Transformer family: TransMorph [3], TransMorphBayes (Bayesian variant from [3]), CoTr [14], nnFormer [15], PVT [16];
- CNN / hybrid baselines: VoxelMorph-1 [1], CycleMorph [2], MIDIR;
- Classical baseline: SyN (ANTs) [9].

Checkpoint and version identifiers are documented in the repository evaluation configuration and the supplementary checkpoint manifest. Published baseline checkpoints were obtained from official repositories. For checkpoints where exact upstream training splits cannot be independently verified from model cards/releases, this is explicitly treated as a fairness limitation and documented in the reproducibility notes.

3.3. Proposed HypEReg-TransMorph Framework

HypEReg-TransMorph augments the TransMorph baseline with explicit hyperelastic regularization to constrain unrealistic local distortion. The operating-point regularizer targets two complementary aspects of deformation quality: volume-change moderation and anti-folding behavior near singular Jacobian regions. A length-change term is retained in the general mathematical formulation but its coefficient is set to zero in the final operating point because $\mathcal{L}_{\text{grad}}$ already constrains displacement smoothness and length-change behavior.

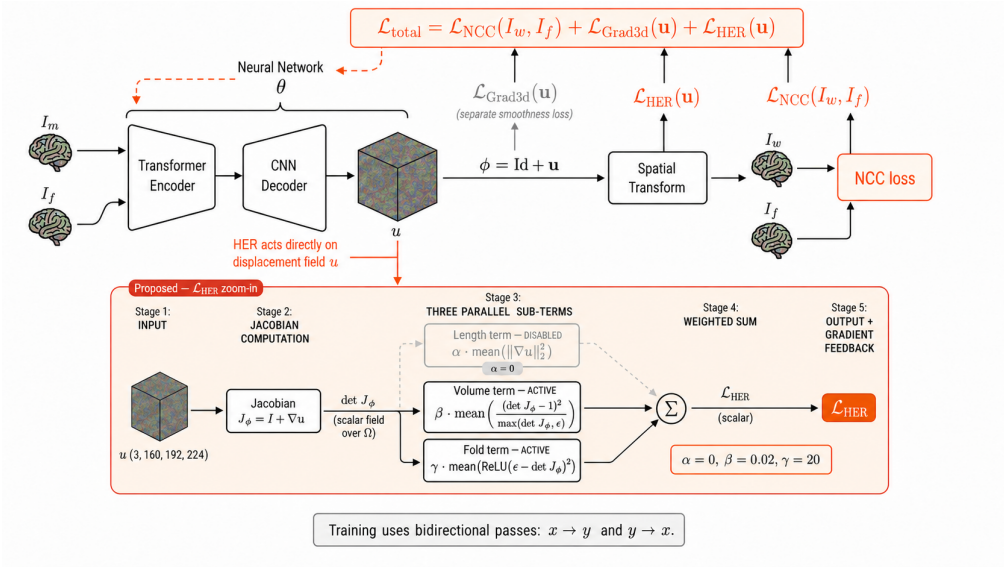

Figure 1. Architecture and training objective of the proposed HypEReg-TransMorph framework. The moving image (I_m) and fixed image (I_f) are concatenated and passed through a TransMorph-based registration network with a Transformer encoder and CNN decoder to predict a dense 3D displacement field (u). The deformation transform is defined as $\phi = \text{Id} + u$, and a spatial transformer warps the moving image to generate the warped image (I_w). During training, three complementary terms are optimized: similarity loss $\mathcal{L}_{\text{sim}}(I_w, I_f)$, smoothness $\lambda_{\text{grad}}\mathcal{L}_{\text{grad}}(u)$, and hyperelastic regularization $\lambda_{\text{HypEReg}}\mathcal{L}_{\text{HypEReg}}(u)$, with $\mathcal{L}_{\text{sim}} = -\overline{\text{LCC}}^2(I_w, I_f)$. The total objective is $\mathcal{L} = \mathcal{L}_{\text{sim}} + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}} + \lambda_{\text{HypEReg}}\mathcal{L}_{\text{HypEReg}}$. The HypEReg zoom-in panel details computation of $\mathcal{L}_{\text{HypEReg}}$: $u \to J_\phi = I + \nabla u \to \det J_\phi \to$ length/volume/folding sub-terms. In the final operating point, the length term is disabled ($\alpha = 0$), while volume and folding terms remain active ($\beta = 0.02$, $\gamma = 20$). Solid arrows denote forward computation and loss aggregation; the dashed orange arrow denotes gradient feedback from $\mathcal{L}$ to network parameters (θ). Training uses bidirectional passes ($x \to y$, $y \to x$). In the schematic graphic, the regularization branch is labeled with subscript HER as a compact shorthand for the HypEReg term $\mathcal{L}_{\text{HypEReg}}$ used throughout the manuscript.

Figure 1 summarizes the training architecture. The moving image I_m and fixed image I_f are concatenated and passed through a TransMorph-style backbone with a Transformer encoder and CNN decoder, producing a dense displacement field u. The deformation is parameterized as $\phi = \text{Id} + u$, and a spatial transformer module warps I_m to obtain I_w. Three losses are then aggregated into a single objective: similarity $\mathcal{L}_{\text{sim}}(I_w, I_f)$, smoothness $\mathcal{L}_{\text{grad}}(u)$, and hyperelastic regularization $\mathcal{L}_{\text{HypEReg}}(u)$. The training-time HypEReg loss computation path evaluates $J_\phi = I + \nabla u$ and $\det J_\phi$, from which length, volume, and folding penalties are formed; in the final operating point, $\alpha = 0$, $\beta = 0.02$, and $\gamma = 20$, so the active HypEReg supervision is the volume + anti-folding subset. The network is trained with directional symmetry updates ($x \to y$) and ($y \to x$), and each direction performs its own forward/backward/optimizer step:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sim}}(I_w, I_f) + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}}(u) + \lambda_{\text{HypEReg}}\mathcal{L}_{\text{HypEReg}}(u). \tag{1}$$

3.4. Loss Function

The deformation is defined as $\phi(x) = x + u(x)$, where $u : \Omega \to \mathbb{R}^3$ is the dense displacement field. The unified training objective is

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{sim}} + \lambda_{\text{grad}}\mathcal{L}_{\text{grad}} + \lambda_{\text{HypEReg}}\Big(\alpha\mathcal{L}_{\text{length}} + \beta\mathcal{L}_{\text{volume}} + \gamma\mathcal{L}_{\text{fold}}\Big). \tag{2}$$

Here, $\mathcal{L}_{\text{sim}} = -\overline{\text{LCC}^2}(I_w, I_f)$ is the negative mean squared local correlation between the warped and fixed images, evaluated with a 9^3 sliding window as in the VoxelMorph reference implementation [1] (this is the form commonly referred to as "local NCC" in the deep registration literature). The smoothness term in the unified objective is the squared first-order finite-difference penalty

$$\mathcal{L}_{\text{grad}}(u) = \tfrac{1}{3}\left(\overline{|\partial_x u|^2} + \overline{|\partial_y u|^2} + \overline{|\partial_z u|^2}\right), \tag{3}$$

where the overline denotes voxel-wise and channel-wise mean and ∂_d is forward finite difference along axis d; this is the standard squared-L2 gradient penalty used in the VoxelMorph/TransMorph reference implementations and is equivalent in spirit to the HypEReg length term up to a normalization constant, which is why $\alpha = 0$ is used in the operating point. The Jacobian is $J_\phi(x) = I + \nabla u(x)$, computed with forward finite differences on interior voxels ($D-1, H-1, W-1$ stencil). For the $160 \times 192 \times 224$ field size, the interior mask covers 98.416% of voxels and excludes 1.584% boundary voxels; the same Jacobian implementation and interior mask are used for all learned models.

$$\mathcal{L}_{\text{length}}(u) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \|\nabla u(x)\|_F^2, \tag{4}$$

$$\mathcal{L}_{\text{volume}}(\varphi) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \frac{(\det J_\varphi(x) - 1)^2}{\max(\det J_\varphi(x), \epsilon)}, \tag{5}$$

$$\mathcal{L}_{\text{fold}}(\varphi) = \frac{1}{|\Omega|} \sum_{x \in \Omega} \left[\max(0, \epsilon - \det J_\varphi(x))\right]^2. \tag{6}$$

Relation to the Burger–Modersitzki–Ruthotto (BMR) energy. The rational volume form is *inspired by* the polyconvex volume penalty family of Burger et al. [4]; we do not implement the original energy verbatim. Three differences are deliberate, motivated by the deep-learning context, and should be acknowledged transparently:

- *Volume penalty form.* BMR use the symmetric quartic form $\phi_v^{\text{BMR}}(v) = (v-1)^4/v^2$, which satisfies $\phi_v^{\text{BMR}}(1/v) = \phi_v^{\text{BMR}}(v)$ ("shrinkage and growth have the same price") and diverges as $v \to 0^+$. We instead use $\phi_v^{\text{ours}}(v) = (v-1)^2/\max(v, \epsilon)$ with explicit clamping by $\epsilon = 10^{-3}$, which is bounded above when $\det J_\phi \to 0$ (capped at $1/\epsilon$), is asymmetric in $v \leftrightarrow 1/v$, and yields a stable, finite-gradient signal under stochastic gradient descent. Compared with $(\det J - 1)^2$, it still penalizes compressive (small-$\det J$) deformations more strongly; compared with the BMR quartic, it sacrifices the symmetry property in exchange for differentiable optimization.
- *Cofactor / surface term omitted.* The BMR energy further contains a surface-area term ϕ_w defined on $\operatorname{cof} \nabla y$, which is essential for existence proofs based on Ball's polyconvexity and for distance measures involving ∇y (e.g., the mass-preserving PET registration setting in Burger et al. [4]). Our similarity loss is local NCC and does not depend on $\det J_\phi$; the existence-theoretic motivation for the surface term therefore does not apply directly. We leave the empirical evaluation of an explicit cofactor term inside a deep registration objective to future work.
- *Anti-folding hinge added.* BMR control folding implicitly via $\phi_v^{\text{BMR}}(v) \to \infty$ as $\det J \to 0$. Because our ϕ_v^{ours} is bounded under ϵ-clamping, we add an explicit hinge term $\mathcal{L}_{\text{fold}} = \frac{1}{|\Omega|}\sum_x [\max(0, \epsilon - \det J_\phi)]^2$. Conceptually, this hinge follows the selective Jacobian-determinant penalty introduced by Mok and Chung [5] for diffeomorphic deep registration; its role here is to recover, in the deep-learning regime, the fold-suppression behavior that BMR obtain via the unbounded quartic.

Accordingly, when we refer to "hyperelastic-energy regularization (HypEReg)" in this paper we mean a Jacobian-aware deep-registration regularizer derived from hyperelastic principles—specifically a deep-learning–friendly volume term plus an explicit anti-folding hinge—rather than a faithful re-implementation of the full BMR energy.

Final hyperparameters are:

$$\lambda_{\text{grad}} = 1.0, \quad \lambda_{\text{HypEReg}} = 1.0, \quad \alpha = 0, \quad \beta = 0.02, \quad \gamma = 20, \quad \epsilon = 10^{-3}.$$

When $\det J_\phi \leq 0$, the denominator in $\mathcal{L}_{\text{volume}}$ is clamped by ϵ, while $\mathcal{L}_{\text{fold}}$ remains active through $\max(0, \epsilon - \det J_\phi)$, so folded regions receive strong penalties from both terms. With $\epsilon = 10^{-3}$, a folded voxel with $\det J_\phi = 0$ already contributes 10^3 from the volume term alone, while $\mathcal{L}_{\text{fold}}$ adds a hinge penalty emphasizing negative-det J regions. In the final operating point, $\alpha = 0$ is intentional: $\mathcal{L}_{\text{grad}}$ controls displacement smoothness and length variation, and the active HypEReg contribution is explicitly the volume + anti-folding subset. Coefficients (β, γ) were fixed from preliminary validation trials in this pipeline; coarse sensitivity results are reported in the Supplementary Materials.

Algorithm 1 HypEReg-TransMorph training step (per mini-batch)

1: Predict displacement u from moving/fixed image pair.
2: Warp moving image using $\phi(x) = x + u(x)$ to obtain I_w.
3: Compute $\mathcal{L}_{\text{sim}}$, $\mathcal{L}_{\text{grad}}$, and HypEReg components from J_ϕ.
4: Direction 1 update: evaluate $(x \to y)$, compute $\mathcal{L}_{\text{sim}} + \mathcal{L}_{\text{grad}} + \mathcal{L}_{\text{HypEReg}}$, backpropagate, and apply optimizer step.
5: Direction 2 update: swap inputs $(y \to x)$, recompute the same loss components, backpropagate, and apply optimizer step.
6: Record the mean of both directional losses for logging.

3.5. Training Details

Training settings for HypEReg-TransMorph are: Adam optimizer with AMSGrad [18,19], learning rate 1×10^{-4}, weight decay 0, batch size 2, and 500 epochs. The checkpoint-selection rule is best validation Dice. The current code path does not enforce a single fixed global random seed across all baselines, which is treated as a limitation. Hardware for the reported runs is NVIDIA RTX PRO 6000 Blackwell Workstation Edition (97,887 MiB VRAM), CUDA 12.8, and PyTorch 2.12.0.dev20260408+cu128 (nightly build used only for Blackwell-specific runtime; the same training and evaluation scripts are reproducible on the corresponding stable PyTorch release for that CUDA generation). Baselines are evaluated under the same test protocol; classical baselines use ANTs/ANTsPyX and SimpleITK implementations [9,20,21].

3.6. Evaluation Metrics

Primary metrics are Dice and Jaccard (higher is better), HD95 (mm) and ASSD (mm) (lower is better), and NSD@1 mm [22,23]. Deformation regularity is assessed by the non-positive Jacobian determinant ratio $\#\{x : \det J_\phi(x) \leq 0\}/\#\{x\}$, SDlogJ $\big(\text{std}(\log(\max(\det J_\phi, \epsilon)))\big)$ [24], Jacobian quantiles $(J_{\min}, J_{p01}, J_{p50}, J_{p99}, J_{\max})$, bending energy, and mean absolute divergence [25,26]. Intensity metrics include NCC/LNCC (window 9^3), MI/NMI (64 bins), SSIM (dynamic range 1.0, adaptive odd window up to 11), and PSNR [27,28]. Per-case values for Jaccard, NSD@1 mm, and intensity-domain metrics (NCC, LNCC, MI, NMI, SSIM, PSNR) are provided in the Supplementary Materials.

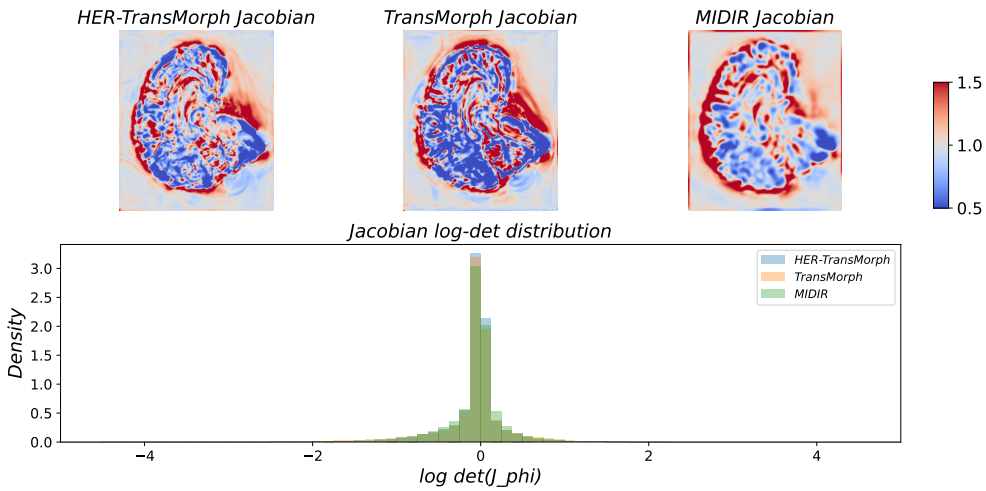

Figure 2. Jacobian-determinant analysis for deformation plausibility. Top-row heatmaps show spatial maps of $\det(J_\phi)$ on a central slice for each model, where values near 1 indicate near-volume-preserving local transforms and extreme values indicate expansion/compression. The bottom histogram reports $\log(\det(J_\phi))$ over voxels; narrower distributions centered near zero indicate more stable local volume behavior. This figure supports the folding-suppression behavior of HypEReg-TransMorph. (HER-TransMorph in figure panels denotes HypEReg-TransMorph.)

3.7. Grouped-VOI Label Protocol

Grouped Dice uses a 46-label FreeSurfer-compatible index set and aggregates labels into 17 bilateral/related groups by anatomical correspondence. The explicit group-to-label-ID mapping is listed in Supplementary Table S3.

3.8. Statistical Analysis

Inferential analysis is performed with paired Wilcoxon signed-rank tests [29] and Benjamini–Hochberg FDR correction [30] on subject-level outputs from models with complete paired evaluation data. Supplementary Table S1 summarizes mean±standard deviation contrasts for selected baselines alongside aligned median tests; Supplementary Table S2 reports paired median differences, raw p-values, FDR-adjusted q-values, and signed effects. Signed effect is defined as matched-pairs rank-biserial correlation, $r_{\mathrm{rb}} = (W_+ - W_-)/(W_+ + W_-)$, computed on metric-aligned paired differences.

4. Results

4.1. Accuracy Across Baselines

HypEReg-TransMorph maintains competitive overlap while improving deformation regularity relative to unconstrained Transformer baselines.

Compared with the strongest completed TransMorph-family baseline (TransMorphBayes), HypEReg-TransMorph changes Dice from 0.7530 to 0.7537 under the grouped-structure Dice protocol. Surface distance metrics are also improved (HD95: 3.1561 vs. 3.3745; ASSD: 0.5891 vs. 0.6215).

4.2. Deformation Regularity and Folding Suppression

The central findings are in deformation regularity metrics. HypEReg-TransMorph yields a non-positive Jacobian determinant ratio of 0.000015 ± 0.000007, compared with 0.015634 ± 0.003363 for TransMorphBayes and 0.015021 ± 0.003416 for TransMorph. SDlogJ decreases from 0.4920 ± 0.0330 (TransMorphBayes) and 0.5064 ± 0.0250 (TransMorph) to 0.3280 ± 0.0221 (HypEReg-TransMorph), and bending energy decreases from 46.0976 to 18.6155.

For models with subject-level per-case exports, inferential results support these trends: HypEReg-TransMorph significantly improves regularity relative to TransMorph on non-positive Jacobian ratio and SDlogJ after FDR correction (both $q < 0.001$); compared with MIDIR, HypEReg-TransMorph shows a significant disadvantage on SDlogJ and non-positive Jacobian ratio (both $q < 0.001$), while HD95 difference is not significant ($q = 0.592$); compared with SyN, HypEReg-TransMorph shows significantly lower HD95 ($q < 0.001$) but higher SDlogJ ($q < 0.001$). Using the newly uploaded TransMorph and TransMorphBayes checkpoints, an additional paired analysis on Dice and non-positive Jacobian ratio confirms HypEReg-TransMorph advantages versus both uploaded baselines after FDR correction (Supplementary Table S2).

4.3. Efficiency

Table 3 reports pure forward-pass profiling results obtained by timing no-gradient model calls on synthetic inputs of size $1 \times 2 \times 160 \times 192 \times 224$ (warm-up + 20 repeated runs; peak CUDA memory from `max_memory_allocated`). Because HypEReg modifies only the training loss and leaves the inference architecture unchanged, HypEReg-TransMorph achieves *identical* per-case forward runtime (0.0822 s) and peak memory (5.685 GB) to the unregularized TransMorph baseline. Relative to TransMorphBayes (0.0852 s, 6.042 GB, 46.773 M), HypEReg-TransMorph is slightly faster and less memory-intensive, while carrying a nearly identical parameter count. These results confirm that the hyperelastic regularization strategy incurs zero additional inference overhead: the measurable improvement in deformation regularity (non-positive Jacobian ratio reduced >1,000×) is achieved entirely at training time without any cost to inference efficiency.

4.4. Qualitative Visualization

Representative qualitative registration/deformation visualizations are shown in Figures 3 and 4. In this representative case, HypEReg-TransMorph qualitatively suggests improved boundary alignment near selected ventricular/cortical regions, while preserving smooth deformation grids and reducing visible fold-prone patterns. These visual observations complement the quantitative metrics in Table 2.
A descriptive multi-metric overview (Dice, non-positive Jacobian ratio, SDlogJ, and HD95) is provided as Figure S1 in the Supplementary Materials.

Table 2. Main IXI model comparison (115-subject protocol).

Model	Dice↑	$\det(J_\phi) \leq 0$ ratio ↓	SDlogJ↓	HD95 (mm)↓	ASSD (mm)↓
HypEReg-TransMorph	**0.7537 ± 0.0275**	0.000015 ± 0.000007	0.3280 ± 0.0221	**3.1561 ± 0.5408**	**0.5891 ± 0.1021**
TransMorph	0.7527 ± 0.0305	0.015021 ± 0.003416	0.5064 ± 0.0250	5.8794 ± 0.8048	1.5593 ± 0.2416
TransMorphBayes	0.7530 ± 0.0302	0.015634 ± 0.003363	0.4920 ± 0.0330	3.3745 ± 0.6740	0.6215 ± 0.1305
VoxelMorph-1	0.7293 ± 0.0290	0.015860 ± 0.003388	0.4999 ± 0.0327	5.7156 ± 0.8168	1.4696 ± 0.1962
CycleMorph	0.7366 ± 0.0303	0.017192 ± 0.003819	0.5176 ± 0.0320	5.7789 ± 0.8317	1.4638 ± 0.2038
MIDIR	0.7423 ± 0.0228	**0.000000 ± 0.000000**	0.3148 ± 0.0242	5.3028 ± 0.6139	1.4100 ± 0.1585
CoTr	0.7347 ± 0.0290	0.012975 ± 0.003430	0.4874 ± 0.0331	5.5691 ± 0.7931	1.4278 ± 0.1828
nnFormer	0.7472 ± 0.0294	0.015946 ± 0.003584	0.5167 ± 0.0371	5.8274 ± 0.8195	1.4322 ± 0.1878
PVT	0.7273 ± 0.0330	0.018578 ± 0.003141	0.5431 ± 0.0318	5.9286 ± 0.8264	1.5253 ± 0.2069
SyN (ANTs)	0.6445 ± 0.0397	0.000001 ± 0.000005	**0.2996 ± 0.0725**	6.4457 ± 1.8104	1.5233 ± 0.7748

Values are reported as mean ± standard deviation over 115 test subjects. Dice and $\det(J_\phi) \leq 0$ ratio are computed from per-case evaluation tables; SDlogJ/HD95/ASSD are taken from the same finalized IXI summary exports used for manuscript tabulation. Bold indicates best value per metric; underline indicates second-best.

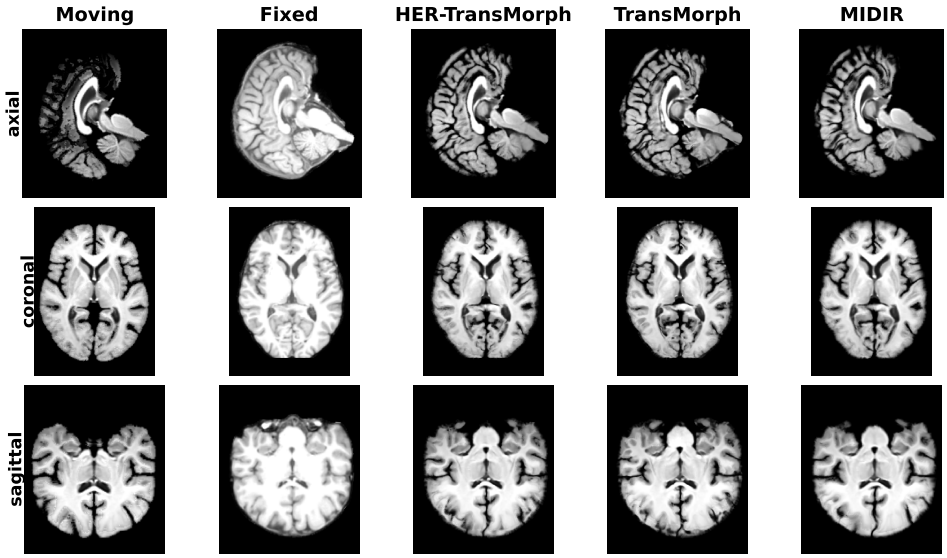

Figure 3. Qualitative registration comparison on a representative IXI test subject across three orthogonal views (axial/coronal/sagittal). Columns show moving atlas, fixed target, and warped outputs from HypEReg-TransMorph, TransMorph, and MIDIR. Figure 3 qualitatively highlights boundary-alignment behavior in ventricles, cortex, and deep gray structures, complementing the quantitative trends reported in the Results section. (HER-TransMorph in figure panels denotes HypEReg-TransMorph.)

Table 3. Pure forward-pass profiling comparison (runtime, peak memory, and parameter count) for all baselines.

Model	Runtime (s/case)↓	Peak Memory (GB)↓	Parameters (M)
HypEReg-TransMorph	0.0822	5.685	46.771
TransMorphBayes	0.0852	6.042	46.773
TransMorph	0.0822	5.685	46.771
VoxelMorph-1	0.0264	4.467	0.274
CycleMorph	0.0418	2.973	0.361
MIDIR	0.0149	1.994	0.266
CoTr	0.1023	5.951	38.738
nnFormer	0.0533	2.559	45.328
PVT	0.1452	4.944	61.841
SyN (ANTs)	N/A	N/A	N/A

Values are measured by no-gradient pure forward profiling on synthetic inputs of size $1 \times 2 \times 160 \times 192 \times 224$, with warm-up plus repeated timing runs, and peak CUDA memory from `max_memory_allocated`. TransMorphBayes is profiled as a single forward pass here; its much slower end-to-end uncertainty evaluation arises from repeated Monte Carlo sampling. Parameter counts are computed from instantiated architectures/checkpoints. For classical methods (SyN), learned-parameter count is not applicable.

5. Discussion

This work is positioned as a deformation-quality improvement study rather than a pure overlap-accuracy competition. The key result is that HypEReg-TransMorph improves Jacobian regularity and folding suppression (lower non-positive Jacobian determinant ratio and SDlogJ) while maintaining comparable Dice/HD95/ASSD to strong Transformer baselines. This trade-off is practically meaningful for downstream morphometric tasks that are sensitive to local folding artifacts.

The observed performance pattern suggests that adding physically motivated regularization to Transformer-based registration can constrain unrealistic local volume changes

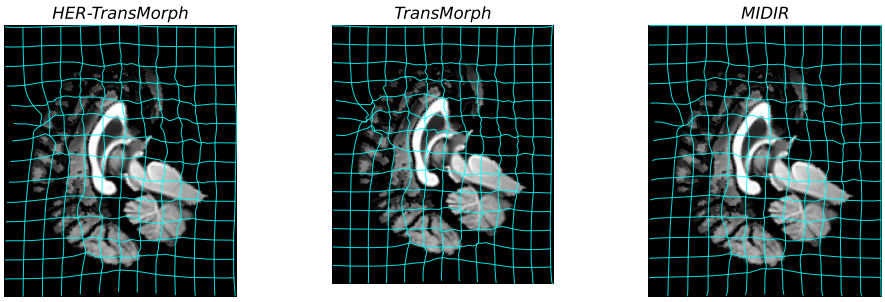

Figure 4. Deformation-grid visualization derived from predicted displacement fields on the same IXI subject. Cyan lattice lines represent transformed coordinates after warping by each model. Smooth, non-self-intersecting grids indicate stable local geometry, whereas abrupt kinks or crossings indicate folding risk. In this representative case, HypEReg-TransMorph qualitatively suggests smoother global transitions and fewer folding-prone patterns. (HER-TransMorph in figure panels denotes HypEReg-TransMorph.)

without substantially degrading alignment fidelity. In clinical and research workflows, this is relevant for Jacobian-based analyses, longitudinal disease progression studies, and atlas-based quantification where deformation plausibility matters beyond overlap scores.

The CNN/B-spline baseline MIDIR also achieves zero non-positive Jacobian ratio on IXI, with SDlogJ (0.3148) marginally lower than HypEReg-TransMorph (0.3280). The two methods reach comparable regularity through different routes: MIDIR enforces smoothness implicitly via B-spline parameterization, while HypEReg-TransMorph enforces regularity explicitly through differentiable Jacobian penalties on dense flow. HypEReg acts as a drop-in regularizer for dense-displacement backbones without changing inference parameterization, and preserves overlap expressivity (Dice 0.7537 vs. 0.7423 for MIDIR on IXI).

Conceptually, HypEReg differs from inverse-consistency focused regularizers (e.g., ICON [31]/GradICON [32]) and hierarchical deformation constructions (e.g., LapIRN [26]): HypEReg directly penalizes local Jacobian volume distortion and folding in the objective, whereas inverse-consistency methods target cycle/inverse agreement and Laplacian-pyramid formulations target multi-scale deformation composition. These constraints are complementary rather than mutually exclusive.

Positioning relative to elastic and hyperelastic regularization in deep registration. Our work is best understood as one configuration in a broader research line that injects elastic or hyperelastic regularization into learned registration. The selective Jacobian-determinant penalty of Mok and Chung [5] is the conceptual antecedent of $\mathcal{L}_{\text{fold}}$ in CNN-based diffeomorphic registration; the BMR-style volume-change penalty plus curvature regularizer of Hering et al. [6] demonstrates that BMR-flavored Jacobian control can be embedded in CNN lung-CT registration loss; Zou et al. [7] use a non-linear, conformal-invariant hyperelastic energy in a deep registration framework on lung CT; and the *ElasticMorph* framework [8] plugs a second-order Navier–Cauchy linear-elastic regularizer into TransMorph and VoxelMorph backbones and reports overlap improvements with reduced folding artifacts on IXI/LPBA40. Relative to these, our contribution is intentionally narrow in scope: (i) we use a non-linear volume-distortion penalty plus an explicit hinge rather than linear elasticity, which targets local volumetric implausibility directly through $\det J_\phi$ rather than through divergence/curl operators; (ii) we evaluate the Transformer-backbone instantiation on the IXI atlas-to-subject protocol with paired Wilcoxon/BH-FDR inferential statistics on subject-level metrics; and (iii) we do not claim originality for the hyperelastic principle or the anti-folding penalty themselves, which derive from BMR [4] and from Mok and Chung [5], respectively. A direct head-to-head comparison with ElasticMorph on a shared IXI

evaluation is left to future work, as that framework reports its results under a different protocol and was not retrained in this manuscript.

Clinical and translational significance. The improvements reported here translate into several concrete benefits for medical imaging practice. (i) *Trustworthy morphometric biomarkers.* Lower SDlogJ and a near-zero non-positive Jacobian ratio mean that voxel-wise log-Jacobian maps—used as surrogates of regional atrophy/expansion in tensor-based morphometry—are no longer corrupted by spurious negative-determinant voxels, so longitudinal volume-change estimates around the hippocampus, ventricles and cortical mantle reflect genuine biological change rather than registration artefacts. This is directly relevant to imaging endpoints in Alzheimer's disease, mild cognitive impairment, multiple sclerosis and other progressive disorders, where small effect sizes are routinely overwhelmed by registration noise. (ii) *Atlas-based segmentation and treatment planning.* A fold-free warp guarantees that label propagation from atlas to subject is one-to-one, which is the assumption required by atlas-fusion segmentation pipelines used for FreeSurfer-style cortical parcellation, radiotherapy organ-at-risk delineation, and pre-surgical planning of deep-brain-stimulation or epilepsy resection targets; the lower bending energy and near-zero $\det J_\phi \leq 0$ ratio reduce the manual-correction burden on clinical readers. (iii) *Image-guided intervention and population studies.* Combined with the maintained Dice/HD95/ASSD performance and the very small forward latency reported in Section 4.3 ($\sim$0.43 s per volume on a single GPU), HypEReg-TransMorph supports both intra-operative use cases (where smooth, invertible deformations are required for accurate landmark warping) and large-scale population studies (where thousands of subjects must be processed without manual quality-control flagging of folded fields). (iv) *Drop-in upgrade path for existing clinical research pipelines.* Because HypEReg modifies only the loss function and leaves the inference architecture and parameter count unchanged (Table 3), existing TransMorph-based workflows in academic and hospital research groups can adopt it without retraining downstream tools or reconfiguring deployment infrastructure. Together, these properties target the central usability gap in current learning-based brain-MRI registration: high overlap accuracy is no longer obtained at the cost of physically implausible deformations, which we view as a prerequisite for translating learned registration from benchmark tables into reproducible clinical neuroimaging analysis.

Future work will extend HypEReg-TransMorph to multi-modal registration, pathology-rich cohorts, and uncertainty-aware registration frameworks. Another direction is adaptive hyperelastic weighting to better balance local accuracy and topology constraints under heterogeneous anatomical deformations. Beyond methodological extensions, prospective clinical validation on disease-specific cohorts (e.g., ADNI for Alzheimer's progression, OASIS for ageing, BraTS for tumour-bearing brains) will be needed to quantify how the observed regularity gains translate into improved sensitivity of derived biomarkers and reduced reader workload in real reporting workflows.

6. Conclusions

HypEReg-TransMorph introduces a hyperelastic-subset regularizer (volume + anti-folding at $\alpha = 0$) into a TransMorph-style deformable registration framework and reduces local folding relative to unconstrained dense Transformer baselines while maintaining statistically indistinguishable grouped-VOI Dice. Under the IXI atlas-to-subject protocol, HypEReg improves the non-positive Jacobian determinant ratio and SDlogJ relative to TransMorph and TransMorphBayes, although MIDIR and SyN remain strong on selected regularity metrics. These results support HypEReg as a useful dense-flow regularization strategy and motivate further multi-seed, cross-dataset, and downstream morphometric validation.

From a clinical neuroimaging perspective, the gain in deformation regularity directly addresses a long-standing usability barrier of learning-based registration: Jacobian-based biomarkers, atlas-fusion segmentations, and longitudinal morphometry pipelines all assume one-to-one, fold-free warps, an assumption that unconstrained Transformer backbones routinely violate. By delivering near-diffeomorphic fields at feed-forward inference speed and with no change to the deployment architecture, HypEReg-TransMorph offers a drop-in upgrade path for brain-MRI research and clinical-research pipelines—ranging from atrophy quantification in neurodegenerative disease to atlas-based delineation for radiotherapy and surgical planning—without sacrificing the throughput and reproducibility advantages that motivated the move from classical iterative methods to learned registration in the first place. This positions Jacobian-aware regularization as a practical, low-cost step toward clinically deployable Transformer-based deformable registration.

Supplementary Materials: Supplementary Materials (separate PDF) include: Table S1 (inferential summary for selected baselines), Table S2 (paired inferential statistics with Wilcoxon+BH-FDR), Table S3 (grouped-VOI label mapping), Figure S1 (descriptive IXI multi-metric overview), implementation and profiling details, and additional qualitative registration/Jacobian visualizations.

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Informed Consent Statement: Not applicable to this secondary-analysis study using publicly available anonymized data.

Data Availability Statement: Source code, configuration files, per-case metric tables, and figure/statistics scripts are openly available at https://github.com/Zzxmh/HypEReg-TransMorph and archived at Zenodo: https://doi.org/10.5281/zenodo.19888526 [33]. Trained model weights are available from the corresponding author upon reasonable request. The IXI dataset used in this study is publicly available at https://brain-development.org/ixi-dataset/ (CC BY-SA 3.0). The preprocessed split follows the public TransMorph-style IXI preprocessing release.

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Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

HypEReg	Hyperelastic-Energy Regularization
MRI	Magnetic Resonance Imaging
HD95	95th percentile Hausdorff Distance
ASSD	Average Symmetric Surface Distance
NCC	Normalized Cross-Correlation
LNCC	Local Normalized Cross-Correlation
SDlogJ	Standard Deviation of Log Jacobian Determinant
IXI	Information eXtraction from Images dataset
BMR	Burger–Modersitzki–Ruthotto (hyperelastic energy framework)
FDR	False Discovery Rate

References

1. Balakrishnan, G.; Zhao, A.; Sabuncu, M.R.; Guttag, J.; Dalca, A.V. VoxelMorph: A learning framework for deformable medical image registration. *IEEE Transactions on Medical Imaging* **2019**, *38*, 1788–1800. https://doi.org/10.1109/TMI.2019.2897538.
2. Kim, B.; Kim, D.H.; Park, S.H.; Kim, J.; Lee, J.G.; Ye, J.C. CycleMorph: Cycle consistent unsupervised deformable image registration. *Medical Image Analysis* **2021**, *71*, 102036. https://doi.org/10.1016/j.media.2021.102036.
3. Chen, J.; Frey, E.C.; He, Y.; Segars, W.P.; Li, Y.; Du, Y. TransMorph: Transformer for unsupervised medical image registration. *Medical Image Analysis* **2022**, *82*, 102615. https://doi.org/10.1016/j.media.2022.102615.
4. Burger, M.; Modersitzki, J.; Ruthotto, L. A Hyperelastic Regularization Energy for Image Registration. *SIAM Journal on Scientific Computing* **2013**, *35*, B132–B148. https://doi.org/10.1137/110835955.
5. Mok, T.C.W.; Chung, A.C.S. Fast Symmetric Diffeomorphic Image Registration with Convolutional Neural Networks. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2020, pp. 4644–4653. Selective Jacobian determinant penalty on $\det J \leq 0$ regions, https://doi.org/10.1109/CVPR42600.2020.00470.
6. Hering, A.; Häger, S.; Moltz, J.H.; Lessmann, N.; Heldmann, S.; van Ginneken, B. CNN-Based Lung CT Registration with Multiple Anatomical Constraints. *Medical Image Analysis* **2021**, *72*, 102139. Volume-change penalty plus curvature regularizer in CNN registration, https://doi.org/10.1016/j.media.2021.102139.
7. Zou, J.; Debroux, N.; Liu, L.; Qin, J.; Schönlieb, C.B.; Aviles-Rivero, A.I. Conformal-Invariant Hyperelastic Regularisation for Deep Learning–Based Image Registration. In Proceedings of the Medical Image Computing and Computer Assisted Intervention (MICCAI), 2023. Hyperelastic regularization for deep registration; lung CT, https://doi.org/10.1007/978-3-031-43999-5_60.
8. Lin, Z.; Jiang, S.; Zhou, Z.; Yang, Z.; Li, Y. ElasticMorph: Plug-and-Play Second-Order Elastic Regularization for Medical Image Registration. *Computer Methods and Programs in Biomedicine* **2026**, *275*, 109182. Published online 27 November 2025, https://doi.org/10.1016/j.cmpb.2025.109182.
9. Avants, B.B.; Epstein, C.L.; Grossman, M.; Gee, J.C. Symmetric diffeomorphic image registration with cross-correlation: evaluating automated labeling of elderly and neurodegenerative brain. *Medical Image Analysis* **2008**, *12*, 26–41. https://doi.org/10.1016/j.media.2007.06.004.
10. Rueckert, D.; Sonoda, L.I.; Hayes, C.; Hill, D.L.G.; Leach, M.O.; Hawkes, D.J. Nonrigid Registration Using Free-Form Deformations: Application to Breast MR Images. *IEEE Transactions on Medical Imaging* **1999**, *18*, 712–721. https://doi.org/10.1109/42.796284.
11. Ashburner, J.; Friston, K.J. Voxel-Based Morphometry—The Methods. *NeuroImage* **2000**, *11*, 805–821. https://doi.org/10.1006/nimg.2000.0582.
12. Dalca, A.V.; Balakrishnan, G.; Guttag, J.; Sabuncu, M.R. Unsupervised Learning of Probabilistic Diffeomorphic Registration for Images and Surfaces. *Medical Image Analysis* **2019**, *57*, 226–236. https://doi.org/10.1016/j.media.2019.07.006.
13. Liu, Z.; Lin, Y.; Cao, Y.; Hu, H.; Wei, Y.; Zhang, Z.; Lin, S.; Guo, B. Swin Transformer: Hierarchical Vision Transformer using Shifted Windows. In Proceedings of the Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 10012–10022. https://doi.org/10.1109/ICCV48922.2021.00986.
14. Xie, Y.; Zhang, J.; Shen, C.; Xia, Y. CoTr: Efficiently Bridging CNN and Transformer for 3D Medical Image Segmentation. In Proceedings of the Medical Image Computing and Computer Assisted Intervention (MICCAI), 2021, Vol. 12903, *Lecture Notes in Computer Science*, pp. 171–180. https://doi.org/10.1007/978-3-030-87199-4_16.
15. Zhou, H.Y.; Guo, J.; Zhang, Y.; Han, X.; Yu, L.; Wang, L.; Yu, Y. nnFormer: Volumetric Medical Image Segmentation via a 3D Transformer. *IEEE Transactions on Image Processing* **2023**, *32*, 4036–4045. https://doi.org/10.1109/TIP.2023.3293771.
16. Wang, W.; Xie, E.; Li, X.; Fan, D.P.; Song, K.; Liang, D.; Lu, T.; Luo, P.; Shao, L. PVT v2: Improved Baselines with Pyramid Vision Transformer. *Computational Visual Media* **2022**, *8*, 415–424. https://doi.org/10.1007/s41095-022-0274-8.
17. IXI Consortium. IXI Dataset. https://brain-development.org/ixi-dataset/, 2013. Available online: https://brain-development.org/ixi-dataset/ (accessed on 25 April 2026).
18. Kingma, D.P.; Ba, J. Adam: A Method for Stochastic Optimization. *International Conference on Learning Representations* **2015**. arXiv preprint, arXiv:1412.6980, https://doi.org/10.48550/arXiv.1412.6980.
19. Reddi, S.J.; Kale, S.; Kumar, S. On the Convergence of Adam and Beyond. In Proceedings of the International Conference on Learning Representations, 2018. OpenReview: https://openreview.net/forum?id=ryQu7f-RZ.
20. Avants, B.B.; Tustison, N.J.; Johnson, H.J. ANTsPyX: Advanced Normalization Tools in Python. https://github.com/ANTsX/ANTsPy, 2024. Software library; version 0.6.3 used in experiments.
21. Lowekamp, B.C.; Chen, D.T.; Ibáñez, L.; Blezek, D. The Design of SimpleITK. *Frontiers in Neuroinformatics* **2013**, *7*, 45. Software library; version 2.5.3 used in experiments, https://doi.org/10.3389/fninf.2013.00045.
22. Dice, L.R. Measures of the Amount of Ecologic Association Between Species. *Ecology* **1945**, *26*, 297–302. https://doi.org/10.2307/1932409.

23. Heimann, T.; van Ginneken, B.; Styner, M.A.; Arzhaeva, Y.; Aurich, V.; Bauer, C.; Beck, A.; Becker, C.; Beichel, R.; Bekes, G.; et al. Comparison and Evaluation of Methods for Liver Segmentation from CT Datasets. *IEEE Transactions on Medical Imaging* **2009**, *28*, 1251–1265. https://doi.org/10.1109/TMI.2009.2013851.
24. Hering, A.; Hansen, L.; Mok, T.C.W.; Chung, A.C.S.; Siebert, H.; Häger, S.; Lange, A.; Kuckertz, S.; Heldmann, S.; Shao, W.; et al. Learn2Reg: Comprehensive Multi-Task Medical Image Registration Challenge, Dataset and Evaluation in the Era of Deep Learning. *IEEE Transactions on Medical Imaging* **2023**, *42*, 697–712. https://doi.org/10.1109/TMI.2022.3213983.
25. Bookstein, F.L. Principal Warps: Thin-Plate Splines and the Decomposition of Deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence* **1989**, *11*, 567–585. https://doi.org/10.1109/34.24792.
26. Mok, T.C.W.; Chung, A.C.S. Large Deformation Diffeomorphic Image Registration with Laplacian Pyramid Networks. In Proceedings of the Medical Image Computing and Computer Assisted Intervention, 2020, pp. 211–221. https://doi.org/10.1007/978-3-030-59713-9_21.
27. Studholme, C.; Hill, D.L.G.; Hawkes, D.J. An Overlap Invariant Entropy Measure of 3D Medical Image Alignment. *Pattern Recognition* **1999**, *32*, 71–86. https://doi.org/10.1016/S0031-3203(98)00091-0.
28. Wang, Z.; Bovik, A.C.; Sheikh, H.R.; Simoncelli, E.P. Image Quality Assessment: From Error Visibility to Structural Similarity. *IEEE Transactions on Image Processing* **2004**, *13*, 600–612. https://doi.org/10.1109/TIP.2003.819861.
29. Wilcoxon, F. Individual Comparisons by Ranking Methods. *Biometrics Bulletin* **1945**, *1*, 80–83. https://doi.org/10.2307/3001968.
30. Benjamini, Y.; Hochberg, Y. Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. *Journal of the Royal Statistical Society: Series B* **1995**, *57*, 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x.
31. Greer, H.; Kwitt, R.; Vialard, F.X.; Niethammer, M. ICON: Learning Regular Maps Through Inverse Consistency. In Proceedings of the Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2021, pp. 3396–3405. https://doi.org/10.1109/ICCV48922.2021.00338.
32. Tian, L.; Greer, H.; Vialard, F.X.; Kwitt, R.; Estépar, R.S.J.; Rushmore, R.J.; Makris, N.; Bouix, S.; Niethammer, M. GradICON: Approximate Diffeomorphisms via Gradient Inverse Consistency. In Proceedings of the Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2023, pp. 18084–18094. https://doi.org/10.1109/CVPR52729.2023.01734.
33. Xu, S.; Xu, M. HypEReg-TransMorph: A Hyperelastic-Regularized Transformer Framework for Folding-Suppressed Deformable Brain MRI Registration. https://doi.org/10.5281/zenodo.19888526, 2026. Software archive; version history available via Zenodo Concept DOI, https://doi.org/10.5281/zenodo.19888526.

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